

These are based on a telescopic system that allows measurement of absolute values of spectral radiance of the light source, without being in contact with it. There are control switches to increase or decrease the number of measurements. In high-speed mode, such cameras provide a measured data in one second to seven seconds.

Integrated light and panel. Usually, with solar LED street-lights, we have LED lights hanging on a pole with a solar panel fitted on top of the pole and a bulky battery connected to the lights. This configuration is quite heavy as well as costly. To tackle this issue, an innovative street-light model was launched this year that has the solar panel integrated with the light. The panel is at the back, light is on the front and battery is inside. The battery gets charged through the panel and then passes the current to the light.

K. Rengarajan, director, Lumino-gloz Solutions Pvt Ltd, says, "As of now, the solar street-lights are very costly, although more power-efficient. Companies are getting into these types of structures for a cost-effective model. If we take an example of 15W integrated solar street-light, it costs around ₹ 10,000 plus taxes, whereas a regular (solar) light costs around ₹ 25,000 plus taxes. While there is a huge difference in the price, the warranty and output on both types of street-lights is the same."

Single-stroke printed circuit board (PCB) printers. We find different types of PCB printers in India, specially for printing PCBs for LED lights. One such printer is a 1.2m street-light PCB printer that uses single-stroke printing technology. The regular surface mount technology (SMT) process includes three steps—printing, pick-n-place and re-flow, and this printer is part of SMT for stencil printing. L. Sampath, managing director, emst Marketing Pvt Ltd, says, "There is a very good market for such printers in India, as there is a huge demand for street-

MAJOR CONTRIBUTORS TO THIS REPORT

- Hemant Jain, honorary secretary general, UCCL
- K. Rengarajan, director, Lumino-gloz Solutions Pvt Ltd
- L. Sampath, managing director, emst Marketing Pvt Ltd
- Praveen Madaan, country head, SMT division, Juki
- Rakesh Shah, director, JSK Innovative Technology Pvt Ltd
- Sumeet Verma, general manager, research and development, Dixon
- Vijay Kumar Gupta, CEO, Kwality Photonics Pvt Ltd



PIR sensor

lights and, in turn, a huge demand for PCBs."

On the other hand, we have a PCB printer that can handle a PCB size of up to 1500mm. This is considered to be a good option for all lights, starting from a tubelight to a small bulb. Another feature of this printer is that it can use all types of components required in the LED light manufacturing process. "The challenge with such machines is maintenance and reliability. However, the best thing about these printing machines is that these are highly reliable and maintenance-free," says Praveen Madaan, country head, SMT division, Juki.

When it comes to LED lighting, these are just some initiatives in terms of innovation. We hope to see more as the market grows.

Current market status and expected growth

The LED lighting industry started almost five years ago in India, when the government policies were not so well-known. If things were different, we would have been at a competitive stage in the market with respect to other countries, including China, Japan and Thailand. However, it is still not too late as the industry is

eager to learn new things. And the industry does not want to sacrifice quality at the cost of price. Rakesh Shah, director, JSK Innovative Technology Pvt Ltd, says, "Today, we have completely indigenous and in-house product de-

velopment, so that it does not hit your pocket and also saves electricity."

On the other hand, Hemant Jain, honourable secretary general, UCCL, says, "With Make in India, quality plays an important role, and we want to project the best quality at a very competitive price." He adds, "Even though the government says that they are helping through the policies, they are actually not. They charge about 14.5 per cent as VAT for LED lights. Such high tax is acceptable when you buy from other countries like China and sell the LED lighting product in India with high margins. Look, if they have to promote it, the tax imposed should either be two per cent or very minimal."

Jain adds, "The government should work on the local eco-system, import duties, taxes, and bring in some subsidies to encourage more and more manufacturing. Currently, import duties on products are about 30-40 per cent."

Whereas, Vijay Kumar Gupta, CEO, Kwality Photonics Pvt Ltd, says, "The demand for LEDs is very encouraging now. It has come to a take-off stage where the government as well as the market is working to take it to a new level. Lately, Energy-Efficiency Services Limited (EESL) has also funded the procurement of LEDs for various municipalities and public entity companies."

Even though everyone has their own take on the market scenario or government policies today, the LED lighting industry as a whole is on the verge of growth and has a long way to go. It is not only improving the quality of products but also attracting more entrepreneurs, which is also increasing the demand for manpower. ●